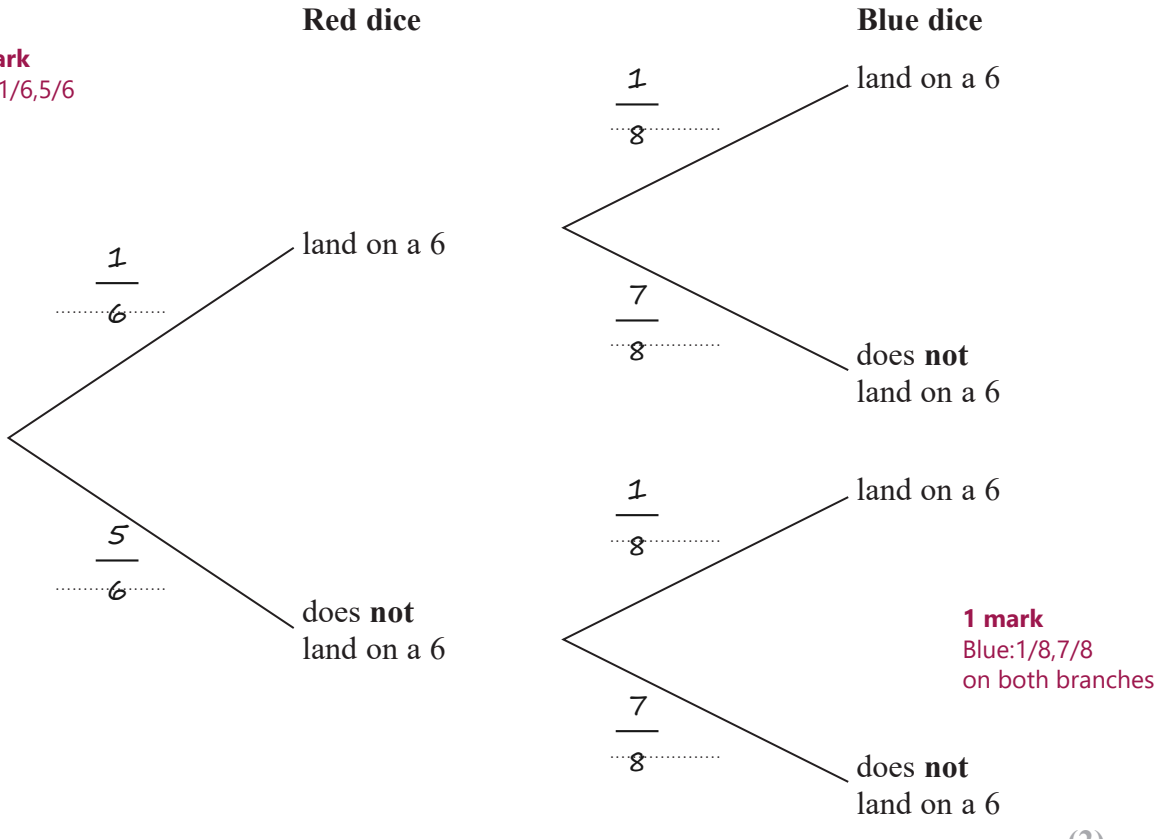
23 Graham has a fair red 6-sided dice and a fair blue 8-sided dice.
The red dice can land on 1, 2, 3, 4, 5 or 6
The blue dice can land on 1, 2, 3, 4, 5, 6, 7 or 8

Graham is going to roll both dice.

(a) Complete the probability tree diagram.



1 mark
Red:1/6,5/6

1 mark
Blue:1/8,7/8
on both branches

(2)

(b) Work out the probability that neither dice will land on a 6

$$P(\text{neither 6}) = \frac{5}{6} \times \frac{7}{8} = \frac{35}{48}$$

1 mark
Multiply the two not6 probabilities

1 mark
35/48

.....

(2)

(Total for Question 23 is 4 marks)